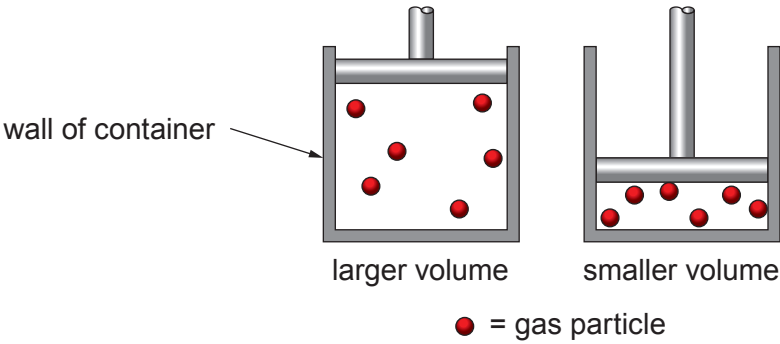


6. Kinetic theory is used to explain the behaviour of gases.

- (a) A fixed mass of gas is kept at a constant temperature in a container. Explain why the pressure increases when its volume is decreased.

[3]



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- (b) Students are given data about the behaviour of a fixed mass of gas kept at **constant pressure**. The volume of the gas changes with temperature as shown in the table below.

Temperature (°C)	Volume (cm ³)
-223	20
-173	40
-73	80



- (i) Chris states that

$$\frac{\text{volume}}{\text{temperature}} = \text{constant}$$

when temperatures are measured in °C. Explain whether the results agree.
Space for calculations.

[2]

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- (ii) The students plot a graph of the results opposite. Explain how they can use the graph to determine the value of absolute zero in °C.

[2]

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- (c) A balloon is filled to a volume of 2800 cm³ at 7 °C (280 K).
The balloon is heated to a temperature of 67 °C.
The pressure remains constant.

Use the equation:

$$\frac{pV}{T} = \text{constant}$$

to calculate the new volume of the balloon.
[$T \text{ (K)} = \theta \text{ (°C)} + 273$]

[4]

volume = cm³

11

